PROJECT-BASED LEARNING

PHASE-II EVALUATION

SkillArc

Team Name: Apex — Team ID: T028

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team Name:	Apex
Team ID:	T028
Course:	DSCPP-III
Semester:	3rd
Domain:	Educational Technology

Team Members

Team Lead	Samiksha Gurung 2510370150
Member	Tanisha Rai 2510370080
Member	Vaibhavi Thapliyal 2510370530

Mentor

Mr. Kuldeep Nautiyal

Phase-I Recap & Feedback Incorporated

Phase-I Proposal

SkillArc was proposed as a gamified learning platform that brings structured roadmaps, study resources, quizzes, coding practice, notes, progress tracking and rewards into one place. The initial scope focuses on DSA and C++ OOP.

Key Needs Identified

- Students need a clear, topic-wise learning path.
- Learning resources and practice are spread across multiple platforms.
- Students need motivation and a way to track progress.

Phase-II Direction

- Organized DSA and OOPS content folders.
- Started a Streamlit-based dashboard and navigation.
- Identified integration, persistence and quiz-navigation tasks .

Problem Statement & Finalized Objectives

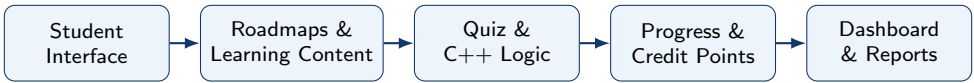
Problem Statement

Students often have to switch between separate websites for learning roadmaps, notes, videos, coding practice and progress tracking. This makes it difficult to follow a structured learning path and maintain consistent practice. SkillArc aims to bring these activities together in one student-focused platform.

Finalized Objectives

1. Provide subject-wise and topic-wise roadmaps for DSA and C++ OOP.
2. Organize notes, learning videos, quizzes and coding-practice resources.
3. Track student activity, topic progress, streaks and Credit Points.
4. Encourage consistency through contests, leaderboards, achievements and planned rewards.

Updated System Architecture / Workflow



Main Components

- DSA topic folders and learning resources
- Streamlit dashboard and sidebar navigation
- C++ quiz prototype with answer validation
- Student progress and Credit Point concepts

Integration Still Required

- Connect interface and quiz/backend modules
- Persist student details, scores and progress
- Complete quiz navigation and final scoring
- Add leaderboard, badges and reward views

Module Design & Functional Requirements

Module	Purpose / Functional Requirement	Phase-II Status
Learning content	Organize DSA topics, notes, README files and video resources.	Initial folders and resources prepared
Student dashboard	Display student information, Credit Points, progress and streaks.	Dashboard prototype started
Quiz system	Display questions, validate answers and update score/Credit Points.	Initial C++ prototype created
Progress storage	Save student details, quiz results and completed topics.	Pending / planned
Rewards & leaderboard	Present rankings, achievements, reward cards and badges.	Pending / planned

Module Dependency

The learning content supplies topics and resources; the quiz module evaluates answers; student data and scores feed into the dashboard. Persistent storage and full integration remain important next steps.

Technology Stack & Development Methodology

Technology Stack

Programming:	C/C++/Python
Frontend Interface:	Streamlit
Storage:	File Handling/Database
Version Control:	Git/GitHub
IDE:	VS Code

Development Methodology

1. Requirement analysis
2. Module and folder structure
3. Individual module development
4. Integration of interface, content and quiz logic
5. Functional testing and validation

Implementation of PBL Course Concepts

Course Concept	Application in SkillArc	Current Evidence
Data Structures in C	Arrays, Stack and Queue topic organization; DSA content is intended to support learning and practice.	Topic folders and basic resources prepared
OOP with C++	`Student` and `Question` classes support the initial quiz prototype.	Question display, answer validation and score/Credit Point updates reported
Python / Streamlit	Student dashboard and sidebar navigation between application sections.	Development environment and dashboard prototype set up
Git / GitHub	Repository structure, collaboration and tracking code changes.	Repository and main branch established
Data persistence	Store student details, scores and progress for future sessions.	Pending; persistent storage planned

Development & Implementation Progress

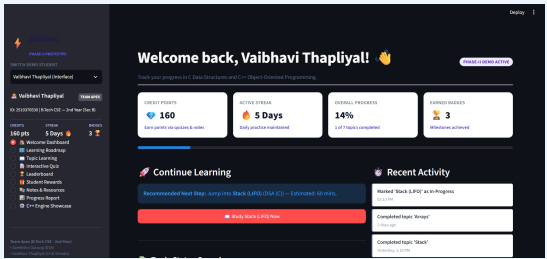
Module	Status	Work Completed / Remaining
DSA content structure	Started	Arrays, Stack and Queue folders, basic notes, README files and YouTube resources prepared.
Student dashboard	In progress	Streamlit environment, dashboard for student information, Credit Points, progress and streaks, and sidebar navigation.
C++ quiz prototype	Initial prototype	Compiler configured; `Student` and `Question` classes, question display, answer validation and score/Credit Point updates reported.
Student registration and storage	Pending	Save student details, quiz results and progress persistently.
Quiz navigation and integration	Pending	Multiple questions, skip/revisit controls, final score calculation and module integration.
Rewards and leaderboard	Pending	Add achievement badges, reward cards and leaderboard views.

Working Prototype / Demonstration

Prototype Elements Reported

- DSA folders for Arrays, Stack and Queue
- Basic notes, README files and learning resources
- Streamlit dashboard and sidebar navigation
- C++ quiz prototype using `Student` and `Question`
- Answer validation and score/Credit Point updates

Prototype Screenshot



Testing, Results & Validation

Test Case	Type	Reported Result / Note	Status
Valid username and name	Positive	Accepts valid characters.	Pass
Invalid username and name	Negative	Report notes invalid characters should be rejected.	Fail
Valid email format	Positive	Accepts .com or .in addresses.	Pass
Invalid email format	Negative	Report notes incorrect formats should be rejected.	Fail
Strong password	Positive	Meets password requirements.	Pass
Weak password	Negative	Password-rule enforcement needs attention.	Fail
Valid login credentials	Positive	Login successful.	Pass
Invalid login credentials	Negative	Error message handling needs attention.	Fail
Empty required fields	Negative	Validation message still needs implementation/testing.	Fail

GitHub Repository & Development Activity

Repository Information

- Repository: Skill-Arc-PBL
- Branch: `main`
- Version Control: Git and GitHub
- Collaboration: Team-based development

Development Activity

- Phase I: Project report and presentation
- Phase II: DSA learning modules and resources
- DSA topics: Arrays, Stacks and Queues
- Project structure maintained through GitHub
- Prototype development and integration

Individual Contribution

Member	Assigned Area	Contribution in Phase-II Report
Samiksha Gurung	DSA content and quiz tasks	Created Arrays, Stack and Queue folders; added basic concept notes, README files and YouTube resources; organized topic folders and prepared the structure for future practice questions and quizzes.
Vaibhavi Thapliyal	Interface and dashboard	Set up Python, Streamlit and GitHub environment; developed the student dashboard for information, Credit Points, progress and streaks; implemented sidebar navigation.
Tanisha Rai	C++ / OOP quiz prototype	Created the project development folders; configured and tested the C++ compiler; created a quiz prototype using `Student` and `Question` classes with question display, answer validation and score/Credit Point updates.

Challenges, Remaining Work & Phase-III Roadmap

Current Challenges

- Integrating DSA content, C++ quiz logic and the interface
- Correctly calculating scores and Credit Points across multiple questions
- Saving student progress after the application closes
- Supporting skip, revisit and smooth quiz navigation
- Keeping the interface clear and easy to use

Phase-III Roadmap

1. Complete registration and persistent student records
2. Expand quizzes with multiple questions and navigation
3. Connect the interface, DSA content and C++ module
4. Add leaderboard, rewards and achievement badges
5. Improve and retest validation and integration
6. Complete documentation and final

Expected Outcomes & References

Expected Final Outcomes

- Structured subject-wise and topic-wise roadmaps
- Centralized notes, videos, quizzes and coding resources
- Student dashboard with progress and Credit Points
- Streaks, leaderboards and achievement tracking
- Peer note sharing and planned monthly rewards

References

1. `https://roadmap.sh/roadmaps`
2. `https://takeuforward.org/home`
3. `https://www.youtube.com/`
4. Yashavant Kanetkar, *Data Structures Through C*, 5th Edition.
5. Herbert Schildt, *C++: The Complete Reference*, 4th Edition.

Thank You

Questions & Suggestions